INCH-POUND

MIL-PRF-27/316C
25 July 2008
SUPERSEDING
MIL-PRF-27/316B
19 May 2003

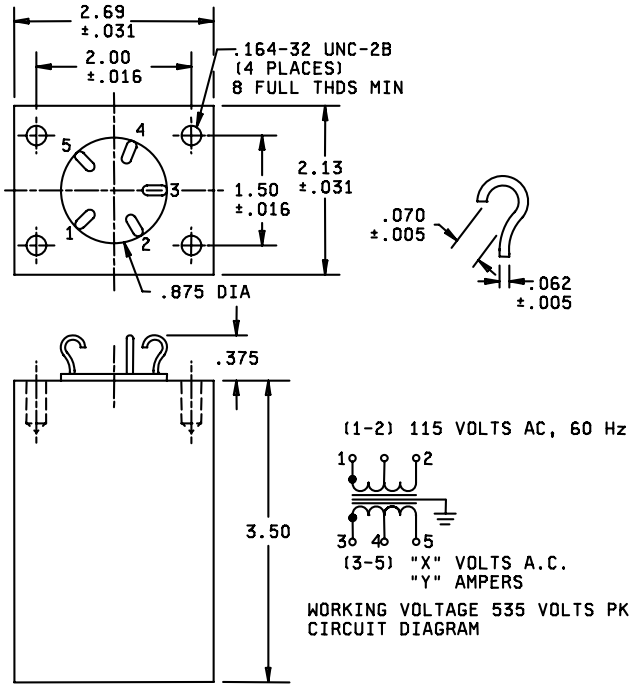
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 30 VOLT-AMPERES, 60 HERTZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.

Inches	mm
.005	0.13
.016	0.41
.031	0.79
.062	1.57
.070	1.78
.375	9.53
.875	22.23
1.50	38.1
2.00	50.8
2.13	54.1
2.69	68.3
3.50	88.9



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the sides of the case.
4. Unless otherwise specified, tolerance shall be ±.063 (1.60 mm).
5. Electrical values shall be marked as specified in table I, except DCR.

FIGURE 1. Dimensions and configurations.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 115 volts ac, 50 to 60 hertz.

Primary dc resistance: 8.5 ohms ±25 percent.

Secondary voltage (3-5): See table I.

Volt-ampere rating: 30 volt-amperes.

Working voltage (1-2): 535 volts peak.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle. Continuous.

Case: Metal enclosed.

Material: Steel.

Terminals: Solderable hook terminals.

Terminal height: 0.375 ±0.063 inch.

Terminal diameter: 0.062 ±0.005 inch.

Weight: 3.0 pounds, maximum.

Operating temperature range. -55° to +130°C.

Altitude: 10,000 feet.

Terminal strength: Method 211, MIL-STD-202, test condition A, 2.0 pounds.

Dielectric withstanding voltage (each winding) (at sea level): 1,500 volts rms.

Electrical characteristics:

Rated load: With 115 volts ac and 60 Hz across (1-2), and rated current in secondary, the voltage across (3-5) shall be as specified in table I.

Regulation: $\frac{Voltage\ (no\ load) - Voltage\ (rated\ load)}{Voltage\ (rated\ load)}\ x\ 100$

Shall not exceed 10 percent.

Polarity: Additive with terminals 2 and 3 connected.

Temperature rise: 35°C with 115 volts ac, 60 hertz across (1-2) at an ambient temperature of 95°C, full load terminals (3-5).

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Marking location: See figure 1.

Part number: M27/316-(dash number from table I).

TABLE I. Electrical ratings.

Dash number	Secondary (3-5)		
	Voltage (volts) ±3 percent	Current 1/ (amperes)	DC resistance (ohms) ±25 percent
01	5.0	6.00	0.04
02	6.3	4.76	0.06
03	8.0	3.76	0.08
04	10.0	3.00	0.14
05	12.6	2.38	0.20
06	14.0	2.15	0.23
07	16.0	1.87	0.28
08	18.0	1.67	0.38
09	20.0	1.50	0.40
10	22.0	1.36	0.55
11	25.0	1.20	0.70
12	28.0	1.07	0.94
13	31.0	0.97	0.96
14	35.0	0.86	1.40
15	39.0	0.77	1.62
16	43.0	0.70	2.15
17	48.0	0.63	2.40
18	55.0	0.55	3.50
19	63.0	0.48	3.90
20	70.0	.429	5.5
21	80.0	.375	6.4
22	90.0	.332	8.6
23	100	.300	10.0
24	110	.272	13.6
25	123	.244	15.3
26	138	.217	21.0
27	157	.191	24.5
28	175	.171	35.0
29	200	.150	40.0
30	220	.136	57.0

1/ For 50 Hz operation, reduce secondary current by 10 percent.

VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/312-26 or M27/316-30 or M27/322-24 shall be sufficient to grant qualification approval to MIL-PRF-27/312 through MIL-PRF-27/322-30, inclusive, all parts.

Qualification testing and approval to M27/316-30 shall be sufficient to grant qualification approval M27/316-01 through M27/316-30.

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents.

MIL-PRF-27
MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2008-043)

Review activities:
Army - CR4, MI
Navy - AS, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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